Uncertainty-Aware Physics-Informed Neural Networks for Incompressible Flow Field Reconstruction: A Comparative Study of Deterministic, MC Dropout, and Split-Head Architectures

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Abstract - Physics-informed neural networks (PINNs) have emerged as a mesh-free alternative for reconstructing flow fields from governing partial differential equations and sparse observations, yet the majority of PINN formulations remain deterministic and provide no indication of predictive confidence. This paper reports a controlled, multi-seed experimental comparison of three PINN variants applied to the reconstruction of a two-dimensional incompressible flow field: a Standard deterministic PINN, an MC Dropout PINN that approximates epistemic uncertainty through stochastic forward passes, and a proposed Split-Head Uncertainty-Quantifying PINN (UQ-PINN) that predicts a heteroscedastic Gaussian output via a shared trunk with separate mean and log-variance heads trained under a Gaussian negative log-likelihood objective. All three architectures were trained and evaluated over five independent random seeds (42, 100, 2026, 777, 999) on an identical spatial domain, and performance was assessed using RMSE, MAE, relative L2 error, negative log-likelihood (NLL), expected calibration error (ECE), and empirical 95% confidence interval coverage. The results reveal a clear and consistent accuracy-uncertainty trade-off: the Standard PINN achieved the lowest streamwise velocity RMSE (0.0426 ± 0.0062) and MAE (0.0238 ± 0.0031), while the Split-Head UQ-PINN produced markedly higher reconstruction error (RMSE_u = 0.2653 ± 0.0419) but the best-calibrated confidence intervals, with mean empirical coverage of 94.74% ± 1.20% against the nominal 95% target, compared to 72.64% ± 2.97% for MC Dropout. The Split-Head model also achieved a more favorable mean NLL (-0.895 versus -0.767 for MC Dropout), although its expected calibration error (0.663) was higher than that of MC Dropout (0.442), indicating that overall confidence-interval coverage and per-bin calibration error do not necessarily move together. A Wilcoxon signed-rank test comparing the Standard PINN and the Split-Head UQ-PINN on streamwise velocity RMSE across the five seeds did not reach statistical significance ($W = 0.0$, $p = 0.0625$, $\alpha = 0.05$), reflecting the limited statistical power of a five-seed sample. These findings indicate that uncertainty-aware PINN architectures can deliver substantially improved and near-nominal confidence-interval coverage relative to MC Dropout, but this benefit is not obtained without a measurable cost in point-prediction accuracy, a trade-off that practitioners must weigh according to application requirements. Index Terms-Physics-informed neural networks, uncertainty quantification, heteroscedastic regression, Monte Carlo dropout, incompressible flow reconstruction, calibration, epistemic uncertainty, aleatoric uncertainty. ## I. INTRODUCTION Reconstructing full flow fields from partial or indirect information is a recurring requirement across fluid mechanics, aerodynamic design, and environmental monitoring. Whether the objective is to infer velocity and pressure distributions from sparse experimental measurements or to accelerate iterative design loops that would otherwise depend on repeated high-fidelity simulation, the underlying computational task is the same: produce a spatially continuous, physically consistent estimate of a flow field given limited data and known governing equations. The reliability of that estimate - not merely its average accuracy but its trustworthiness at any given point in the domain - is what ultimately determines whether such reconstructions can be used for downstream engineering decisions. Conventional computational fluid dynamics (CFD) techniques, built on finite volume, finite element, or finite difference discretizations, remain the workhorse of high-fidelity flow analysis. These methods are mathematically well understood and can achieve high accuracy when the mesh, boundary conditions, and numerical scheme are appropriately chosen. However, they carry practical costs that become significant in iterative or data-assimilation contexts: mesh generation for complex geometries is labor-intensive, mesh quality directly constrains solution accuracy, and each new geometry, boundary condition, or parameter setting typically requires a fresh simulation. When the goal is not to solve the forward problem from scratch but to reconstruct a field consistent with both governing physics and sparse observational data, mesh-based solvers do not naturally accommodate a hybrid physics-plus-data objective without substantial reformulation, such as data assimilation or adjoint-based approaches that add further computational overhead. Physics-informed neural networks (PINNs) offer an alternative formulation in which the flow field is represented as a continuous function approximated by a neural network, and the governing partial differential equations (PDEs) are enforced as soft constraints via automatic differentiation rather than through explicit discretization. This mesh-free character allows PINNs to fuse governing physics with sparse or noisy observational data within a single unified training objective, which is attractive for reconstruction problems where full-field ground truth is unavailable. Since their formalization, PINNs have been applied to a wide range of forward and inverse problems in fluid mechanics, heat transfer, and structural mechanics, demonstrating that a trained network can act as a compact, differentiable surrogate for the governing PDE system. Despite this promise, the overwhelming majority of PINN implementations remain deterministic: for a given spatial query, the network returns a single point estimate with no accompanying measure of confidence. This is a significant limitation in any setting where the reconstructed field will inform a decision, because a deterministic PINN offers no mechanism to distinguish a well-constrained prediction, supported by dense data and low physics residual, from a poorly constrained one, extrapolated far from any observation or physics anchor. In engineering practice, knowing that a prediction is uncertain is often as valuable as the prediction itself, since it identifies where additional sensors, simulation refinement, or caution is warranted. Predictive uncertainty in machine learning is typically decomposed into two components. Epistemic uncertainty reflects the model's own lack of knowledge - uncertainty that could in principle be reduced with more data or a better model - and is often associated with regions of sparse observational coverage. Aleatoric uncertainty, by contrast, reflects irreducible noise or variability inherent in the observations themselves and does not vanish even with infinite data. A physically meaningful uncertainty-aware PINN should ideally be capable of representing both. Monte Carlo (MC) Dropout is one of the most widely adopted approaches for approximating epistemic uncertainty in deep networks without requiring an explicit Bayesian formulation. By keeping dropout active at inference time and performing repeated stochastic forward passes, the resulting distribution of predictions is used as an approximate posterior, from which a predictive mean and variance can be estimated. MC Dropout is architecturally simple and can be layered onto an existing PINN with minimal modification, but its variance estimate is driven primarily by the randomness of the dropout mask rather than by an explicit model of data noise, and its calibration properties in physics-constrained settings are not guaranteed. An alternative strategy is to model the aleatoric component of uncertainty directly, through a heteroscedastic output formulation in which the network predicts not only a mean but also an input-dependent variance, trained under a Gaussian negative log-likelihood (NLL) objective. This "split-head" design allows the network to learn where the data itself is noisy or under-constrained and to widen its predictive intervals accordingly, in principle producing calibration behavior that differs qualitatively from the dropout-based epistemic approximation. The research gap addressed in this paper is the absence of a controlled, multi-seed, side-by-side comparison of these three PINN formulations - Standard, MC Dropout, and Split-Head heteroscedastic - evaluated not only on point-prediction accuracy but on the full suite of uncertainty-relevant metrics: negative log-likelihood, expected calibration error, and empirical confidence-interval coverage. Much of the existing PINN uncertainty literature reports either accuracy or calibration in isolation, or evaluates a single random seed, leaving open the question of whether reported differences reflect genuine architectural effects or seed-to-seed variability. The objective of this study is to provide exactly this controlled comparison. We train all three architectures on an identical two-dimensional flow reconstruction problem, using five independent random seeds per model, and report the full distribution of results rather than a single run. The contributions of this paper, grounded directly in the experimental evidence obtained, are as follows. First, we quantify the accuracy cost of uncertainty-aware training relative to a deterministic baseline, showing that the Standard PINN achieves substantially lower RMSE and MAE than either uncertainty-aware variant. Second, we show that the Split-Head UQ-PINN achieves markedly better empirical 95% confidence-interval coverage (94.74%) than MC Dropout (72.64%), bringing coverage close to the nominal target. Third, we report a more favorable mean NLL for the Split-Head model, alongside a higher expected calibration error than MC Dropout, revealing that different calibration diagnostics can disagree and must be interpreted jointly rather than in isolation. Fourth, we apply a Wilcoxon signed-rank test across the five seeds and report that the observed RMSE difference between the Standard PINN and the Split-Head UQ-PINN, while numerically large, does not reach conventional statistical significance at this sample size, and we discuss this limitation candidly rather than overstating the strength of the evidence. ## II. RELATED WORK The formal development of physics-informed neural networks as a general framework for solving and inferring solutions to nonlinear PDEs established the now-standard practice of embedding governing equation residuals directly into the training loss via automatic differentiation, allowing a single network to satisfy both data-fitting and physics-consistency objectives simultaneously. This formulation proved attractive because it removes the need for mesh generation and allows physical constraints to be imposed at arbitrary collocation points in space and time, rather than at fixed discretization nodes. Subsequent research extended PINNs into fluid mechanics specifically, applying the framework to incompressible Navier-Stokes problems for both forward simulation and inverse inference of unknown flow quantities from sparse velocity or scalar transport observations. This body of work demonstrated that PINNs could recover pressure fields that are not directly observed, by exploiting the coupling between pressure and velocity enforced through the momentum equations, and that reconstruction quality depends strongly on the density and placement of collocation points and observational data. These fluid-mechanics-focused PINN studies, however, are almost universally deterministic in their output formulation, producing a single scalar estimate per query point without any accompanying uncertainty measure. The broader deep learning literature on uncertainty quantification offers a substantially more developed toolkit, most of which has only partially migrated into the PINN setting. Bayesian neural networks, in principle, offer a fully probabilistic treatment of network weights, but exact Bayesian inference is intractable for networks of practical size, motivating approximate methods. MC Dropout was proposed as a computationally efficient approximation to Bayesian inference, in which dropout layers, ordinarily used only during training as a regularizer, are kept active during inference and multiple stochastic forward passes are aggregated to estimate a predictive distribution. This approach requires no change to the training objective and can be retrofitted onto an existing deterministic architecture, which has made it a popular default choice for uncertainty-aware extensions of PINNs. However, several studies outside the PINN literature have noted that MC Dropout variance is heavily influenced by the dropout rate and network capacity rather than by the actual structure of the data-generating process, and that its calibration is not guaranteed to match the nominal confidence level associated with the chosen interval. Heteroscedastic regression, in which a network predicts both a mean and an input-dependent variance and is trained under a Gaussian negative log-likelihood objective, offers a complementary and largely orthogonal approach that targets aleatoric rather than epistemic uncertainty. This approach has been used extensively in general-purpose deep learning uncertainty estimation, where it is often combined with ensembling or dropout to jointly capture both uncertainty types. Its appeal in a physics-informed setting is that it allows the network to explicitly learn where observational data is noisy or physically under-constrained and widen its predictive interval accordingly, rather than relying on the essentially data-independent stochasticity of a dropout mask. Calibration and reliability analysis -

quantifying whether a model's stated confidence intervals match observed outcome frequencies - has a long history in probabilistic forecasting and has become a standard diagnostic in modern deep learning uncertainty quantification, typically reported through expected calibration error (ECE) computed from reliability diagrams, or through empirical coverage of nominal confidence intervals. These calibration metrics are not always mutually consistent: a model can achieve favorable aggregate coverage while still displaying miscalibration within specific confidence bins, since coverage is a single-number summary of interval performance whereas ECE evaluates the full calibration curve across bins. This distinction is directly relevant to the results reported in the present study. Within the specific intersection of physics-informed learning and uncertainty quantification, prior work has proposed both Bayesian PINN formulations and dropout-based or ensemble-based PINN variants, generally reporting that uncertainty-aware training changes the accuracy profile of the network relative to a purely deterministic baseline. However, direct multi-seed comparisons that report accuracy, calibration, and coverage together for MC Dropout and heteroscedastic split-head PINNs evaluated under identical training and evaluation protocols remain comparatively rare, which is the gap this paper addresses through controlled experimentation. ## III. PROBLEM FORMULATION AND MATHEMATICAL BACKGROUND ### A. Governing Physical Equations The reconstruction task addressed in this study concerns a steady, two-dimensional, incompressible viscous flow defined over a rectangular spatial domain spanning the streamwise coordinate x \in [1, 8] and the transverse coordinate y \in [-2, 2], as illustrated by the collocation point distribution in Fig. 1. The flow state at any point in the domain is described by the streamwise velocity component u(x,y), the transverse velocity component v(x,y), and the pressure field p(x,y). The flow is governed by the incompressible Navier-Stokes equations, comprising the continuity equation enforcing mass conservation, \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{1} and the steady-state momentum conservation equations in the streamwise and transverse directions, u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right), \tag{2} u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right), \tag{3} where Re denotes the Reynolds number of the non-dimensionalized flow. These equations express, respectively, that the fluid is divergence-free and that momentum is conserved subject to pressure gradients and viscous diffusion. ### B. Physics Residual Formulation In the PINN framework, the field variables u, v, and p are represented as outputs of a neural network f_\theta(x,y) parameterized by weights \theta. Rather than discretizing (1)-(3) on a mesh, the governing equations are enforced by evaluating their residuals at a set of collocation points distributed throughout the domain, using automatic differentiation to compute the required partial derivatives of the network output with respect to its spatial inputs. Defining the continuity residual as r_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \tag{4} and the momentum residuals as r_u = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \frac{1}{Re}\nabla^2 u, \tag{5} r_v = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \frac{1}{Re}\nabla^2 v, \tag{6} training drives these residuals toward zero at every sampled collocation point, so that the network's output becomes an approximate solution of the governing PDE system across the domain, without requiring the equations to be satisfied only at discrete mesh nodes. ### C. Loss Function Formulation The total training objective for a deterministic PINN is a weighted sum of a data-fidelity term and a physics-residual term, \mathcal{L} = \lambda_d \mathcal{L}{data} + \lambda_p \mathcal{L}{physics}, \tag{7} where \mathcal{L}{data} penalizes the discrepancy between network predictions and available observations, typically as a mean squared error, and \mathcal{L}{physics} penalizes the squared residuals of (4)-(6) evaluated at the collocation points. The scalar weights \lambda_d and \lambda_p balance the relative influence of data fidelity and physical consistency during optimization. For the heteroscedastic Split-Head formulation used in this study, the data-fidelity term is replaced by a Gaussian negative log-likelihood that incorporates a predicted variance \sigma^2(x,y) alongside the predicted mean \hat{u}(x,y), \mathcal{L}{NLL} = \frac{1}{N} \sum{i=1}^{N}\left[\frac{(u_i - \hat{u}_i)^2}{2\sigma_i^2} + \frac{1}{2}\log \sigma_i^2\right], \tag{8} where the network predicts the log-variance \log \sigma_i^2 directly for numerical stability, and N is the number of observed data points used at each training step. This loss allows the network to attenuate the penalty on data points where it predicts a larger variance, at the cost of a corresponding regularization term that discourages the variance from growing without bound. ## IV. PROPOSED METHODOLOGY ### A. Standard PINN The Standard PINN serves as the deterministic baseline against which the two uncertainty-aware architectures are compared. It receives the spatial coordinates (x, y) as input and outputs a single deterministic estimate of the flow state (\hat{u}, \hat{v}, \hat{p}) through a fully connected feed-forward network. Training minimizes the composite loss in (7), combining a mean-squared-error data loss against available observations with the physics-residual loss derived from (4)-(6). Because the network architecture and forward pass are entirely deterministic, a single query at any spatial location returns exactly one prediction with no accompanying measure of confidence; the model provides no coverage statistic and no calibration behavior to report, which is reflected in the 0.0% coverage value recorded for this model in the experimental results. ### B. MC Dropout PINN The MC Dropout PINN retains the same physics-informed training objective as the Standard PINN but incorporates dropout layers within the network trunk. Ordinarily, dropout is active only during training as a stochastic regularizer and is disabled at inference. In the MC Dropout formulation, dropout remains active at inference time as well, so that each forward pass through the network produces a slightly different output due to the randomly sampled dropout mask. By performing T stochastic forward passes at a given query point and treating the resulting set of predictions as samples from an approximate predictive distribution, a predictive mean \bar{u}(x,y) = \frac{1}{T}\sum_{t=1}^{T} \hat{u}_t(x,y) and predictive variance \text{Var}[\hat{u}(x,y)] can be estimated empirically across the T samples. This variance is interpreted as an approximation of epistemic uncertainty, reflecting the model's own confidence given its current parameterization, and is used to construct approximate 95% confidence intervals around the predictive mean under a Gaussian assumption on the sample distribution. ### C. Split-Head UQ-PINN The Split-Head UQ-PINN shares a common trunk of hidden layers with the other architectures but diverges into two separate output heads following the shared representation: a mean-prediction head that outputs $\hat{u}(x,y)$, $\hat{v}(x,y)$, and $\hat{p}(x,y)$, and a variance-prediction head that outputs the corresponding log-variance $\log \sigma^2(x,y)$ for each predicted quantity. This architecture is trained under the Gaussian NLL objective given in (8), jointly with the physics-residual loss enforcing (4)-(6), so that the physics constraint continues to shape the mean prediction while the variance head learns, from the data-fidelity term alone, where the model's predictions should be treated as more or less certain. Because the variance head is trained end-to-end alongside the mean head and the physics residual, its output is intended to reflect aleatoric uncertainty - variability attributable to the structure of the observational data and the degree to which the physics constraint is locally well-posed - rather than the sampling-based epistemic approximation produced by MC Dropout. At inference, the predicted variance from this head is used directly to construct 95% confidence intervals without requiring repeated stochastic forward passes. ### D. Aleatoric and Epistemic Uncertainty The two uncertainty-aware architectures in this study are designed to target conceptually distinct sources of predictive uncertainty. The MC Dropout model estimates epistemic uncertainty by treating the spread of predictions across stochastic forward passes as a proxy for the model's own parameter uncertainty; this quantity is expected to be higher where the model is less constrained by data or physics and could, in principle, be reduced with additional training data or model capacity. The Split-Head model's variance head is trained to estimate aleatoric uncertainty, representing irreducible variability that would persist even for a perfectly specified model, learned directly from the mismatch between predictions and observations during optimization of the Gaussian NLL objective. Fig. 8 presents the epistemic uncertainty map obtained from the MC Dropout model and Fig. 9 presents the aleatoric uncertainty map obtained from the Split-Head model, allowing a qualitative comparison of the spatial structure - or lack thereof - associated with each uncertainty type. ### E. Training and Inference Procedure All three architectures were trained on the same spatial domain and collocation configuration described in Section V, using the composite data-and-physics loss objective appropriate to each model. Training and validation loss were tracked over 2,000 training iterations (Fig. 2), with both curves decreasing on a logarithmic scale from an initial value near 5×10^{-2} to a final value below 2×10^{-3}, and the validation loss tracking the training loss closely throughout, remaining marginally above it at all recorded iterations without diverging. At inference, the Standard PINN produces a single deterministic field, the MC Dropout PINN produces a predictive mean and variance from repeated stochastic forward passes, and the Split-Head UQ-PINN produces a predictive mean and variance directly from its two output heads in a single forward pass. Beyond the number of training iterations and the general architecture of each model, this study does not report additional training hyperparameters (e.g., learning rate schedule, optimizer settings, exact network width or depth, or dropout rate) because these were not present in the supplied experimental configuration; readers requiring exact reproduction should refer to the accompanying implementation. ## V. EXPERIMENTAL SETUP ### A. Computational Domain The evaluation domain is a two-dimensional rectangle spanning $x \in [1, 8]$ in the streamwise direction and $y \in [-2, 2]$ in the transverse direction, consistent across all ground-truth and predicted-field figures (Figs. 3-11). This domain is sampled with a dense, regular grid of evaluation points visible in the scatter pattern of Figs. 3-11, providing the spatial resolution at which each model's predictions were compared against ground truth. ### B. Data and Collocation Configuration Collocation points used to enforce the physics residual loss were sampled along two vertical boundary lines of the domain, at approximately $x \approx 1.0$ and $x \approx 4.55$, each spanning the full transverse extent $y \in [-2, 2]$, as shown in Fig. 1. This configuration indicates that the physics constraint was enforced along these two spatial cross-sections rather than through a fully dense two-dimensional collocation grid across the entire rectangular domain, meaning the interior of the domain away from these two lines relies more heavily on the combination of the physics-residual propagation through the network and any available data observations than on directly sampled collocation points in that region. ### C. Experimental Models Three PINN variants were evaluated under an otherwise identical training and evaluation protocol: the Standard PINN (deterministic baseline), the MC Dropout PINN (stochastic-inference epistemic uncertainty), and the Split-Head UQ-PINN (heteroscedastic Gaussian aleatoric uncertainty). All three models share the same underlying physics-residual formulation described in Section III and were evaluated on the same held-out spatial grid. ### D. Multi-Seed Evaluation Protocol Each of the three models was independently trained and evaluated across five random seeds: 42, 100, 2026, 777, and 999. This multi-seed protocol allows the reported performance metrics to be summarized as a mean and standard deviation across seeds, rather than as a single-run result, and enables a paired statistical significance test (Section VI-J) comparing models on a per-seed basis. All quantitative results reported in Section VI are computed directly from the seed-level values in the supplied performance records unless otherwise noted. ### E. Evaluation Metrics Model performance was assessed using root-mean-square error (RMSE) and mean absolute error (MAE) for the streamwise (u) and transverse (v) velocity components, relative L2 error for the streamwise velocity, negative log-likelihood (NLL) for the two probabilistic models, expected calibration error (ECE) derived from the reliability diagram in Fig. 12, and empirical coverage of the nominal 95% confidence interval. Because the Standard PINN produces no predictive distribution, NLL, ECE, and non-zero coverage are not defined for this model and are reported as such rather than estimated or approximated. ## VI. RESULTS ### A. Training and Validation Convergence Fig. 2 shows the training and validation loss curves recorded over 2,000 iterations on a logarithmic vertical axis. Both curves decrease smoothly and monotonically from an initial value close to 5×10^{-2} at iteration zero to values near 8×10^{-4} (training) and 1.3×10^{-3} (validation) by iteration 2,000. The validation loss remains consistently slightly above the training loss throughout the full training window, and the gap between the two curves does not widen appreciably in the later stages of training, which is consistent with stable convergence and does not, on its own, indicate overfitting within the recorded iteration range. ### B. Streamwise Velocity Reconstruction Figs. 3 through 6 present, respectively, the ground-truth streamwise velocity field and the predictions of the Standard PINN, the MC Dropout PINN, and the Split-Head UQ-PINN. All four fields share the same domain and color scale, spanning approximately -0.2 to 1.3. The ground-truth field (Fig. 3) exhibits a low-velocity, near-zero to negative recirculation region concentrated near x \in [1, 3], y \in [-0.7, 0.7], flanked above and below by higher-velocity regions exceeding u \approx 1.2 near the inlet boundary, with the flow generally accelerating and becoming more spatially uniform toward the outlet at x = 8. Both the Standard PINN (Fig. 4) and the Split-Head UQ-PINN (Fig. 6) reproduce this qualitative structure closely, capturing the low-velocity recirculation pocket near the inlet and the high-velocity bands above and below it, with the predicted fields visually near-indistinguishable from the ground truth at this resolution. The MC Dropout PINN prediction (Fig. 5) also reproduces the overall structure but shows a visibly less sharp transition around the low-velocity core near x \approx 1-3 and a somewhat broader, less concentrated high-velocity band near y \approx 1-1.5 compared to the ground truth and the other two models, consistent with the elevated quantitative RMSE and MAE reported for this model below. Quantitatively, the mean streamwise velocity RMSE across five seeds was 0.0426 \pm 0.0062 for the Standard PINN, 0.1252 \pm 0.0132 for MC Dropout, and 0.2653 \pm 0.0419 for the Split-Head UQ-PINN, an ordering that places the Standard PINN as the most accurate point predictor of the three, MC Dropout intermediate, and the Split-Head model as the least accurate on this metric. The corresponding mean absolute errors follow the same ordering: 0.0238 \pm 0.0031 (Standard), 0.0878 \pm 0.0085 (MC Dropout), and 0.1985 \pm 0.0305 (Split-Head). ### C. Spatial Reconstruction Error Fig. 7 shows the spatial distribution of absolute error, with values ranging up to approximately 0.002 across the domain according to the associated color scale. The error field displays a spatially diffuse, speckled pattern without a clearly concentrated region of elevated error; small, scattered pockets of higher error (approaching the upper end of the displayed color scale) and lower error (near the dark end of the scale) are interspersed throughout the domain rather than clustering systematically near the inlet recirculation region or the outlet boundary. This visual pattern indicates that, at the resolution and color scale depicted in Fig. 7, reconstruction error is not strongly localized to any single sub-region of the flow field; we do not treat the specific numeric magnitudes suggested by the colorbar as equivalent to the aggregate RMSE/MAE statistics reported in Section VI-B, which were computed directly from the underlying performance data rather than derived from the figure. ### D. Transverse Velocity Reconstruction Fig. 10 shows the reconstructed transverse velocity field v(x,y), using a diverging color scale centered near zero and ranging from approximately -0.6 to 0.5. The field exhibits an alternating pattern of positive and negative transverse velocity lobes distributed along the streamwise direction: a region of strongly negative v centered near x \approx 2-3, y \approx 0, adjacent to a region of strongly positive v centered near x \approx 4-5, y \approx 0, followed by a second region of negative v near x \approx 6-7.5. This alternating structure is consistent with the presence of a recirculating flow feature near the inlet, since transverse velocity components of opposing sign on either side of a recirculation zone are a typical signature of such flow structures. The field decays toward near-zero transverse velocity at the upper and lower edges of the domain (y \to \pm 2), consistent with the flow becoming more streamwise-aligned away from the central recirculation region. ### E. Pressure Field Reconstruction Fig. 11 presents the reconstructed pressure field p(x,y), which shows a smooth, low-frequency spatial variation without the sharper localized features present in the velocity fields. Pressure is lowest (most negative, approaching -0.5) in a compact region near the inlet at approximately x \approx 1-2, y \approx 0-0.5, coincident with the location of the low-velocity recirculation feature identified in the streamwise velocity fields, and increases smoothly and monotonically with streamwise distance toward the outlet, where it plateaus near zero across most of the transverse extent. This qualitative pattern - a pressure minimum co-located with the recirculation zone and a smooth pressure recovery downstream - is physically consistent with the expected coupling between the low-velocity core and a locally reduced pressure region, though no separate quantitative pressure-error metric was available in the supplied performance data, and we accordingly restrict this discussion to the qualitative spatial pattern visible in Fig. 11. ### F. Epistemic and Aleatoric Uncertainty Fig. 8 presents the epistemic uncertainty map obtained from the MC Dropout model, and Fig. 9 presents the aleatoric uncertainty map obtained from the Split-Head model's variance head. Both maps use similarly scaled color ranges (approximately 0 to 0.005-0.006) and display a broadly speckled, spatially non-uniform pattern across the domain without an obvious systematic trend tied to the recirculation zone or domain boundaries. Neither uncertainty map shows a clearly interpretable spatial gradient of the kind that would, for example, indicate elevated epistemic uncertainty concentrated specifically near the two collocation lines shown in Fig. 1 or specifically far from them; the patterns in both Fig. 8 and Fig. 9 appear largely stochastic at the resolution presented, and we do not attribute a stronger spatial interpretation to these maps than the figures directly support. ### G. Reliability and Calibration Fig. 12 presents the reliability diagram used to compute expected calibration error. The plotted calibration curve tracks the diagonal identity line (representing perfect calibration) closely across most of the confidence range from 0 to approximately 0.7, with the curve lying marginally above the diagonal in the upper-middle range (roughly 0.6-0.9 predicted confidence), indicating a modest tendency toward overconfidence in that band, before converging back toward the diagonal near the highest confidence bin. The quantitative expected calibration error computed from this and analogous per-model reliability data was 0.4420 \pm 0.0339 for MC Dropout and 0.6631 \pm 0.0616 for the Split-Head UQ-PINN, indicating that, judged strictly by this ECE metric, MC Dropout exhibited lower (better) calibration error than the Split-Head model across the five seeds - a result that stands in contrast to the coverage-based comparison discussed next, and one we return to directly in the Discussion. ### H. 95% Confidence Interval Coverage Fig. 13 compares empirical 95% confidence interval coverage between MC Dropout and Split-Head UQ-PINN. The mean empirical coverage across five seeds was 72.64% \pm 2.97% for MC Dropout and 94.74% \pm 1.20% for the Split-Head UQ-PINN, against a nominal target of 95%. The Split-Head model's coverage is close to the nominal target and displays a lower standard deviation across seeds than MC Dropout, whose coverage falls substantially short of the 95% target and displays greater seed-to-seed variability. The Standard PINN, being fully deterministic, does not produce a confidence interval and its coverage is recorded as 0.0% with zero variance across all five seeds, consistent with the absence of any predictive distribution for this model.

I. Multi-Seed Quantitative Evaluation

Table III summarizes the mean and standard deviation of the principal accuracy metrics across the five evaluated seeds for each model. Table IV summarizes the corresponding uncertainty and calibration metrics.

TABLE III. MULTI-SEED PERFORMANCE COMPARISON (MEAN ± SD, N = 5 SEEDS)

Model	RMSE_u	RMSE_v	MAE_u	Rel. L2_u
Standard PINN	0.0426 ± 0.0062	0.0384 ± 0.0052	0.0238 ± 0.0031	0.0471 ± 0.0069
MC Dropout PINN	0.1252 ± 0.0132	0.1246 ± 0.0139	0.0878 ± 0.0085	0.1383 ± 0.0146
Split-Head UQ-PINN	0.2653 ± 0.0419	0.2748 ± 0.0006	0.1985 ± 0.0305	0.2931 ± 0.0463

Lower values indicate better point-prediction accuracy. Bold indicates the lowest (best) value in each column.

TABLE IV. UNCERTAINTY AND CALIBRATION METRICS (MEAN ± SD, N = 5 SEEDS)

Model	NLL	ECE	Coverage (%)
Standard PINN	N/A	N/A	0.00 ± 0.00
MC Dropout PINN	-0.767 ± 0.434	0.442 ± 0.034	72.64 ± 2.97
Split-Head UQ-PINN	-0.895 ± 0.228	0.663 ± 0.062	94.74 ± 1.20

For NLL, a lower (more negative) value is generally favorable. For ECE, lower is better. For coverage, proximity to the nominal 95% target is the relevant criterion. Bold indicates the more favorable value per metric; the Standard PINN produces no predictive distribution and is not applicable (N/A) for these columns.

The RMSE_u standard deviation for the Standard PINN (0.0062) is smaller in absolute terms than that of the Split-Head model (0.0419), and the coefficient of variation (std/mean) is broadly comparable in relative terms across the three models, indicating that seed-to-seed variability scales roughly with the magnitude of each model's error rather than one model being disproportionately unstable relative to its own performance level.

J. Statistical Significance Analysis

A Wilcoxon signed-rank test was applied to compare the Standard PINN and the Split-Head UQ-PINN on paired, per-seed streamwise velocity RMSE values across the five common seeds. The test statistic was W = 0.0 with an associated p-value of 0.0625. Table V reports this result.

TABLE V. STATISTICAL SIGNIFICANCE TEST RESULTS

Comparison	Test	Statistic	p-value	Significant at \alpha=0.05
Standard PINN vs. Split-Head UQ-PINN (RMSE_u)	Wilcoxon signed-rank	0.0	0.0625	No

At a conventional significance threshold of \alpha = 0.05, this result does not achieve statistical significance (p = 0.0625 > 0.05), even though the Wilcoxon statistic of W = 0.0 indicates that the Standard PINN achieved a lower RMSE than the Split-Head UQ-PINN on every one of the five paired seeds - the maximally consistent possible outcome for a five-sample paired test. This apparent tension is a direct consequence of the small sample size: with only five paired observations, the Wilcoxon signed-rank test cannot reach a p-value below 0.05 even under a perfectly consistent directional result, because the minimum achievable p-value for n=5 paired non-tied samples is exactly 0.0625. We report this result transparently rather than characterizing a directionally consistent but formally non-significant outcome as statistically established.

VII. DISCUSSION

A. Accuracy-Uncertainty Trade-Off

The central empirical finding of this study is a pronounced accuracy-uncertainty trade-off across the three evaluated architectures. The Standard PINN, unconstrained by any requirement to model predictive variance, achieved the lowest RMSE and MAE for streamwise velocity by a substantial margin - roughly three to six times lower error than the two uncertainty-aware variants. A plausible explanation for this gap is that both uncertainty-aware models must allocate representational and optimization capacity to a secondary objective - either the implicit regularization introduced by dropout-based stochasticity in the MC Dropout case, or the explicit variance-head parameterization and Gaussian NLL loss in the Split-Head case - and this additional objective may compete with, rather than purely complement, the primary mean-prediction accuracy objective during training. This interpretation is consistent with, though not directly proven by, the ordering of RMSE values observed here, and should be treated as a candidate mechanism rather than an established causal account.

B. Calibration Behavior

The coverage results in Section VI-H and the ECE results in Section VI-G present what might initially appear to be a contradictory picture: the Split-Head UQ-PINN achieves coverage (94.74%) much closer to the nominal 95% target than MC Dropout (72.64%), yet simultaneously exhibits a higher (worse) ECE (0.663) than MC Dropout (0.442). This is not a logical inconsistency but a reflection of the fact that these two metrics probe different aspects of calibration. Empirical coverage is a single aggregate statistic answering the question "does the stated 95% interval actually contain the true value roughly 95% of the time, on average across the whole domain?" ECE, computed from a reliability diagram with multiple confidence bins (Fig. 12), instead asks whether predicted confidence at every level of confidence - not just the 95% level - matches observed accuracy at that level. A model could achieve strong aggregate 95% coverage while still being poorly calibrated at other confidence levels, for instance if its predictive distribution is well-scaled overall but has a distorted shape relative to a true Gaussian, which would inflate ECE without necessarily harming coverage at the specific 95% threshold. The results reported here are consistent with this possibility for the Split-Head model, though we do not have direct evidence to isolate the exact source of the elevated ECE and note this as an open question. Meanwhile, MC Dropout's comparatively low ECE alongside its markedly poor 95% coverage may indicate that its predictive variance, while reasonably well-scaled on average across confidence bins, is nonetheless systematically too narrow specifically at high confidence levels, a pattern that would be consistent with the widely reported tendency of MC Dropout to underestimate predictive uncertainty relative to the true generalization error, since dropout-induced variance is driven by network stochasticity rather than a direct model of prediction error.

C. Model-Level Interpretation

The NLL results provide an additional, distinct perspective. The Split-Head UQ-PINN achieved a more favorable (more negative) mean NLL (-0.895) than MC

Dropout (-0.767), despite having substantially higher point-prediction error. This is possible because NLL jointly rewards both prediction accuracy and appropriately scaled uncertainty - a model that is less accurate but appropriately widens its predictive interval to reflect that reduced accuracy can achieve a comparable or better NLL than a more accurate model whose intervals are poorly scaled relative to its actual error. This pattern could be attributed to the Split-Head model's explicit training under the Gaussian NLL objective in (8), which directly couples the predicted variance to the observed residual during optimization, whereas MC Dropout's variance emerges only indirectly from the stochasticity of the dropout mask and is not directly optimized against a likelihood-based objective. The qualitative uncertainty maps in Figs. 8 and 9 do not, on visual inspection, reveal an obvious spatial pattern that would further explain this difference, and we do not draw a stronger conclusion from those two figures than the diffuse, speckled patterns they display. ### D. Practical Implications Taken together, these results suggest that the choice between a deterministic PINN, an MC Dropout PINN, and a heteroscedastic Split-Head UQ-PINN should be guided by the specific requirements of the downstream application rather than by a single "best" model across all metrics. Where the primary requirement is the most accurate possible point estimate of the flow field and predictive uncertainty is not needed, the Standard PINN's substantially lower RMSE and MAE make it the preferred choice based on this experiment. Where downstream decisions depend on having a reliable, near-nominal confidence interval - for instance, flagging regions of the domain where a prediction should not be trusted without further verification - the Split-Head UQ-PINN's coverage result close to the nominal 95% target is a meaningful practical advantage over MC Dropout, whose intervals in this experiment covered the true value only about 73% of the time despite being nominally constructed as 95% intervals, a level of miscalibration that could be operationally consequential if such intervals were relied upon for safety-relevant or resource-allocation decisions. At the same time, the higher ECE of the Split-Head model indicates that its calibration is not uniformly strong across all confidence levels, and practitioners intending to use predictive probabilities directly (rather than solely a single nominal interval) should be aware of this limitation. ## VIII. LIMITATIONS This study has several limitations that should be considered when interpreting its findings. First, and most significantly, the statistical comparisons in this paper are based on only five random seeds per model. While five seeds are sufficient to compute a mean and standard deviation and to observe a consistent directional trend, they provide limited statistical power, as illustrated directly by the Wilcoxon test result in Section VI-J, where a perfectly consistent five-out-of-five directional outcome still could not reach the conventional \alpha = 0.05 significance threshold. Any claims of "improvement" or "degradation" between models in this paper should be read as observations from a limited-sample experimental comparison rather than as statistically confirmed population-level effects. Second, the evaluation is restricted to a single two-dimensional incompressible flow reconstruction problem on a fixed spatial domain and a fixed collocation configuration (two vertical lines, as shown in Fig. 1). The generalizability of the observed accuracy-uncertainty trade-off to other flow regimes, three-dimensional domains, unsteady flows, or different collocation strategies has not been tested in this study and should not be assumed. Third, quantitative pressure-field error metrics were not available in the supplied experimental results; the discussion of Fig. 11 in Section VI-E is therefore necessarily qualitative, and we have deliberately avoided assigning a numerical accuracy value to the pressure reconstruction that was not directly measured. Fourth, the uncertainty modeling assumptions underlying both uncertainty-aware architectures carry their own limitations: MC Dropout's variance estimate depends on the specific dropout configuration used during training, which was not detailed in the supplied experimental configuration, and the Split-Head model's Gaussian NLL formulation assumes a Gaussian predictive distribution, which may not fully capture the true error distribution if it is skewed or heavy-tailed. Fifth, this paper reports only the metrics and figures supplied in the accompanying experimental results; certain training hyperparameters (optimizer settings, learning rate schedule, exact network width and depth, dropout rate, and total wall-clock or GPU-hour cost per model) were not present in the supplied material and are therefore not reported here, which may limit exact reproducibility by readers without access to the underlying implementation. Finally, the spatial uncertainty maps in Figs. 8 and 9 were interpreted only at the qualitative, visual level supported by the figures; more rigorous spatial statistical analysis of these uncertainty fields - for example, formal spatial autocorrelation testing - was outside the scope of the supplied experimental results and was not attempted here. ## IX. CONCLUSION AND FUTURE WORK This paper presented a controlled, five-seed experimental comparison of three physics-informed neural network architectures - a Standard deterministic PINN, an MC Dropout PINN, and a Split-Head heteroscedastic UQ-PINN - applied to the reconstruction of a two-dimensional incompressible flow field. The experiment demonstrated a clear and internally consistent accuracy-uncertainty trade-off: the Standard PINN achieved the lowest streamwise velocity RMSE (0.0426 ± 0.0062) and MAE (0.0238 ± 0.0031) of the three models and is the strongest choice when point-prediction accuracy alone is the objective, while the Split-Head UQ-PINN achieved the most favorable mean NLL (-0.895 ± 0.228) and the empirical 95% confidence-interval coverage closest to the nominal target (94.74% ± 1.20%, versus 72.64% ± 2.97% for MC Dropout), making it the stronger choice when reliable predictive uncertainty is the priority. This same experiment also produced a result that complicates a simple "Split-Head is best for uncertainty" narrative: MC Dropout achieved a lower (better) expected calibration error (0.442 ± 0.034) than the Split-Head model (0.663 ± 0.062), showing that different calibration diagnostics can favor different models and that no single uncertainty-aware architecture dominated across every uncertainty metric evaluated. A Wilcoxon signed-rank test comparing Standard PINN and Split-Head UQ-PINN RMSE across the five seeds did not reach statistical significance at \alpha = 0.05 (p = 0.0625), despite a perfectly consistent directional advantage for the Standard PINN across all five paired seeds, underscoring the limited statistical power available at this sample size. The central implication for practitioners selecting among uncertainty-aware PINN architectures is that no model evaluated in this study was uniformly superior: the choice between a deterministic baseline, an MC Dropout extension, and a heteroscedastic Split-Head formulation should be driven by whether

the priority is minimizing point-prediction error, maximizing the reliability of a specific nominal confidence interval, or maintaining low calibration error across the full range of predicted confidence levels, since these three criteria were not simultaneously optimized by any single model in this experiment. Future work motivated directly by the limitations identified in Section VIII should prioritize increasing the number of independent random seeds per model to improve statistical power beyond what a five-seed Wilcoxon test can support, extending the evaluation to additional flow regimes and three-dimensional or unsteady domains to test the generalizability of the observed accuracy-uncertainty trade-off, and incorporating quantitative pressure-field error metrics alongside the currently available velocity-based metrics. Further work could also investigate hybrid architectures that combine the Split-Head heteroscedastic variance head with MC Dropout stochastic inference within a single network, in order to jointly model aleatoric and epistemic uncertainty and evaluate whether such a combination can achieve both near-nominal coverage and lower expected calibration error than either individual approach achieved in this study. Finally, a systematic ablation over collocation point density and placement - extending beyond the two-line configuration shown in Fig. 1 - would help clarify whether the observed spatial uncertainty patterns in Figs. 8 and 9 are influenced by proximity to collocation regions, a question this experiment's figures could not resolve on their own. ##
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I. INTRODUCTION

Reconstructing full flow fields from partial or indirect information is a recurring requirement across fluid mechanics, aerodynamic design, and environmental monitoring. Whether the objective is to infer velocity and pressure distributions from sparse experimental measurements or to accelerate iterative design loops that would otherwise depend on repeated high-fidelity simulation, the underlying computational task is the same: produce a spatially continuous, physically consistent estimate of a flow field given limited data and known governing equations. The reliability of that estimate - not merely its average accuracy but its trustworthiness at any given point in the domain - is what ultimately determines whether such reconstructions can be used for downstream engineering decisions.

Conventional computational fluid dynamics (CFD) techniques, built on finite volume, finite element, or finite difference discretizations, remain the workhorse of high-fidelity flow analysis. These methods are mathematically well understood and can achieve high accuracy when the mesh, boundary conditions, and numerical scheme are appropriately chosen. However, they carry practical costs that become significant in iterative or data-assimilation contexts: mesh generation for complex geometries is labor-intensive, mesh quality directly constrains solution accuracy, and each new geometry, boundary condition, or parameter setting typically requires a fresh simulation. When the goal is not to solve the forward problem from scratch but to reconstruct a field consistent with both governing physics and sparse observational data, mesh-based solvers do not naturally accommodate a hybrid physics-plus-data objective without substantial reformulation, such as data assimilation or adjoint-based approaches that add further computational overhead.

Physics-informed neural networks (PINNs) offer an alternative formulation in which the flow field is represented as a continuous function approximated by a neural network, and the governing partial differential equations (PDEs) are enforced as soft constraints via automatic differentiation rather than through explicit discretization. This mesh-free character allows PINNs to fuse governing physics with sparse or noisy observational data within a single unified training objective, which is attractive for reconstruction problems where full-field ground truth is unavailable. Since their formalization, PINNs have been applied to a wide range of forward and inverse problems in fluid mechanics, heat transfer, and structural mechanics, demonstrating that a trained network can act as a compact, differentiable surrogate for the governing PDE system.

Despite this promise, the overwhelming majority of PINN implementations remain deterministic: for a given spatial query, the network returns a single point estimate with no accompanying measure of confidence. This is a significant limitation in any setting where the reconstructed field will inform a decision, because a deterministic PINN offers no mechanism to distinguish a well-constrained prediction, supported by dense data and low physics residual, from a poorly constrained one, extrapolated far from any observation or physics anchor. In engineering practice, knowing that a prediction is uncertain is often as valuable as the prediction itself, since it identifies where additional sensors, simulation refinement, or caution is warranted.

Predictive uncertainty in machine learning is typically decomposed into two components. Epistemic uncertainty reflects the model's own lack of knowledge - uncertainty that could in principle be reduced with more data or a better model - and is often associated with regions of sparse observational coverage. Aleatoric uncertainty, by contrast, reflects irreducible noise or variability inherent in the observations themselves and does not vanish even with infinite data. A physically meaningful uncertainty-aware PINN should ideally be capable of representing both.

Monte Carlo (MC) Dropout is one of the most widely adopted approaches for approximating epistemic uncertainty in deep networks without requiring an explicit Bayesian formulation. By keeping dropout active at inference time and performing repeated stochastic forward passes, the resulting distribution of predictions is used as an approximate posterior, from which a predictive mean and variance can be estimated. MC Dropout is architecturally simple and can be layered onto an existing PINN with minimal modification, but its variance estimate is driven primarily by the randomness of the dropout mask rather than by an explicit model of data noise, and its calibration properties in physics-constrained settings are not guaranteed.

An alternative strategy is to model the aleatoric component of uncertainty directly, through a heteroscedastic output formulation in which the network predicts not only a mean but also an input-dependent variance, trained under a Gaussian negative log-likelihood (NLL) objective. This "split-head" design allows the network to learn where the data itself is noisy or under-constrained and to widen its predictive intervals accordingly, in principle producing calibration behavior that differs qualitatively from the dropout-based epistemic approximation.

The research gap addressed in this paper is the absence of a controlled, multi-seed, side-by-side comparison of these three PINN formulations - Standard, MC Dropout, and Split-Head heteroscedastic - evaluated not only on point-prediction accuracy but on the full suite of uncertainty-relevant metrics: negative log-likelihood, expected calibration error, and empirical confidence-interval coverage. Much of the existing PINN uncertainty literature reports either accuracy or calibration in isolation, or evaluates a single random seed, leaving open the question of whether reported differences reflect genuine architectural effects or seed-to-seed variability.

The objective of this study is to provide exactly this controlled comparison. We train all three architectures on an identical two-dimensional flow reconstruction problem, using five independent random seeds per model, and report the full distribution of results rather than a single run. The contributions of this paper, grounded directly in the experimental evidence obtained, are as follows. First, we quantify the accuracy cost of uncertainty-aware training relative to a deterministic baseline, showing that the Standard PINN achieves substantially lower RMSE and MAE than either uncertainty-aware variant. Second, we show that the Split-Head UQ-PINN achieves markedly better empirical 95% confidence-interval coverage (94.74%) than MC Dropout (72.64%), bringing coverage close to the nominal target. Third, we report a more favorable mean NLL for the Split-Head model, alongside a higher expected calibration error than MC Dropout, revealing that different calibration diagnostics can disagree and must be interpreted jointly rather than in isolation. Fourth, we apply a Wilcoxon signed-rank test across the five seeds and report that the observed RMSE difference between the Standard PINN and the Split-Head UQ-PINN, while numerically large, does not reach conventional statistical significance at this sample size, and we discuss this limitation candidly rather than overstating the strength of the evidence.

II. RELATED WORK

The formal development of physics-informed neural networks as a general framework for solving and inferring solutions to nonlinear PDEs established the now-standard practice of embedding governing equation residuals directly into the training loss via automatic differentiation, allowing a single network to satisfy both data-fitting and physics-consistency objectives simultaneously. This formulation proved attractive because it removes the need for mesh generation and allows physical constraints to be imposed at arbitrary collocation points in space and time, rather than at fixed discretization nodes.

Subsequent research extended PINNs into fluid mechanics specifically, applying the framework to incompressible Navier-Stokes problems for both forward simulation and inverse inference of unknown flow quantities from sparse velocity or scalar transport observations. This body of work demonstrated that PINNs could recover pressure fields that are not directly observed, by exploiting the coupling between pressure and velocity enforced through the momentum equations, and that reconstruction quality depends strongly on the density and placement of collocation points and observational data. These fluid-mechanics-focused PINN studies, however, are almost universally deterministic in their output formulation, producing a single scalar estimate per query point without any accompanying uncertainty measure.

The broader deep learning literature on uncertainty quantification offers a substantially more developed toolkit, most of which has only partially migrated into the PINN setting. Bayesian neural networks, in principle, offer a fully probabilistic treatment of network weights, but exact Bayesian inference is intractable for networks of practical size, motivating approximate methods. MC Dropout was proposed as a computationally efficient approximation to Bayesian inference, in which dropout layers, ordinarily used only during training as a regularizer, are kept active during inference and multiple stochastic forward passes are aggregated to estimate a predictive distribution. This approach requires no change to the training objective and can be retrofitted onto an existing deterministic architecture, which has made it a popular default choice for uncertainty-aware extensions of PINNs. However, several studies outside the PINN literature have noted that MC Dropout variance is heavily influenced by the dropout rate and network capacity rather than by the actual structure of the data-generating process, and that its calibration is not guaranteed to match the nominal confidence level associated with the chosen interval.

Heteroscedastic regression, in which a network predicts both a mean and an input-dependent variance and is trained under a Gaussian negative log-likelihood objective, offers a complementary and largely orthogonal approach that targets aleatoric rather than epistemic uncertainty. This approach has been used extensively in general-purpose deep learning uncertainty estimation, where it is often combined with ensembling or dropout to jointly capture both uncertainty types. Its appeal in a physics-informed setting is that it allows the network to explicitly learn where observational data is noisy or physically under-constrained and widen its predictive interval accordingly, rather than relying on the essentially data-independent stochasticity of a dropout mask.

Calibration and reliability analysis - quantifying whether a model's stated confidence intervals match observed outcome frequencies - has a long history in probabilistic forecasting and has become a standard diagnostic in modern deep learning uncertainty quantification, typically reported through expected calibration error (ECE) computed from reliability diagrams, or through empirical coverage of nominal confidence intervals. These calibration metrics are not always mutually consistent: a model can achieve favorable aggregate coverage while still displaying miscalibration within specific confidence bins, since coverage is a single-number summary of interval performance whereas ECE evaluates the full calibration curve across bins. This distinction is directly relevant to the results reported in the present study.

Within the specific intersection of physics-informed learning and uncertainty quantification, prior work has proposed both Bayesian PINN formulations and dropout-based or ensemble-based PINN variants, generally reporting that uncertainty-aware training changes the accuracy profile of the network relative to a purely deterministic baseline. However, direct multi-seed comparisons that report accuracy, calibration, and coverage together for MC Dropout and heteroscedastic split-head PINNs evaluated under identical training and evaluation protocols remain comparatively rare, which is the gap this paper addresses through controlled experimentation.

III. PROBLEM FORMULATION AND MATHEMATICAL BACKGROUND

A. Governing Physical Equations

The reconstruction task addressed in this study concerns a steady, two-dimensional, incompressible viscous flow defined over a rectangular spatial domain spanning the streamwise coordinate $x \in [1, 8]$ and the transverse coordinate $y \in [-2, 2]$, as illustrated by the collocation point distribution in Fig. 1. The flow state at any point in the domain is described by the streamwise velocity component $u(x,y)$, the transverse velocity component $v(x,y)$, and the pressure field $p(x,y)$.

The flow is governed by the incompressible Navier-Stokes equations, comprising the continuity equation enforcing mass conservation,

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \tag{1}$$

and the steady-state momentum conservation equations in the streamwise and transverse directions,

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right), \tag{2}$$

$$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right), \tag{3}$$

where Re denotes the Reynolds number of the non-dimensionalized flow. These equations express, respectively, that the fluid is divergence-free and that momentum is conserved subject to pressure gradients and viscous diffusion.

B. Physics Residual Formulation

In the PINN framework, the field variables u, v, and p are represented as outputs of a neural network $f_\theta(x,y)$ parameterized by weights θ. Rather than discretizing (1)-(3) on a mesh, the governing equations are enforced by evaluating their residuals at a set of collocation points distributed throughout the domain, using automatic differentiation to compute the required partial derivatives of the network output with respect to its spatial inputs. Defining the continuity residual as

$$r_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \tag{4}$$

and the momentum residuals as

$$r_u = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \frac{1}{Re}\nabla^2 u, \tag{5}$$

$$r_v = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \frac{1}{Re}\nabla^2 v, \tag{6}$$

training drives these residuals toward zero at every sampled collocation point, so that the network's output becomes an approximate solution of the governing PDE system across the domain, without requiring the equations to be satisfied only at discrete mesh nodes.

C. Loss Function Formulation

The total training objective for a deterministic PINN is a weighted sum of a data-fidelity term and a physics-residual term,

$$\mathcal{L} = \lambda_d \mathcal{L}{data} + \lambda_p \mathcal{L}{physics}, \tag{7}$$

where $\mathcal{L}{data}$ penalizes the discrepancy between network predictions and available observations, typically as a mean squared error, and $\mathcal{L}{physics}$ penalizes the squared residuals of (4)-(6) evaluated at the collocation points. The scalar weights λ_d and λ_p balance the relative influence of data fidelity and physical consistency during optimization.

For the heteroscedastic Split-Head formulation used in this study, the data-fidelity term is replaced by a Gaussian negative log-likelihood that incorporates a predicted variance $\sigma^2(x,y)$ alongside the predicted mean $\hat{u}(x,y)$,

$$\mathcal{L}{NLL} = \frac{1}{N}\sum{i=1}^{N}\left[\frac{(u_i - \hat{u}_i)^2}{2\sigma_i^2} + \frac{1}{2}\log \sigma_i^2\right], \tag{8}$$

where the network predicts the log-variance $\log \sigma_i^2$ directly for numerical stability, and N is the number of observed data points used at each training step. This loss allows the network to attenuate the penalty on data points where it predicts a larger variance, at the cost of a corresponding regularization term that discourages the variance from growing without bound.

IV. PROPOSED METHODOLOGY

A. Standard PINN

The Standard PINN serves as the deterministic baseline against which the two uncertainty-aware architectures are compared. It receives the spatial coordinates (x, y) as input and outputs a single deterministic estimate of the flow state ($\hat{u}$, $\hat{v}$, $\hat{p}$) through a fully connected feed-forward network. Training minimizes the composite loss in (7), combining a mean-squared-error data loss against available observations with the physics-residual loss derived from (4)-(6). Because the network architecture and forward pass are entirely deterministic, a single query at any spatial location returns exactly one prediction with no accompanying measure of confidence; the model provides no coverage statistic and no calibration behavior to report, which is reflected in the 0.0% coverage value recorded for this model in the experimental results.

B. MC Dropout PINN

The MC Dropout PINN retains the same physics-informed training objective as the Standard PINN but incorporates dropout layers within the network trunk. Ordinarily, dropout is active only during training as a stochastic regularizer and is disabled at inference. In the MC Dropout formulation, dropout remains active at inference time as well, so that each forward pass through the network produces a slightly different output due to the randomly sampled dropout mask. By performing T stochastic forward passes at a given query point and treating the resulting set of predictions as samples from an approximate predictive distribution, a predictive mean $\bar{u}(x,y) = \frac{1}{T} \sum_{t=1}^{T} \hat{u}_t(x,y)$ and predictive variance $\text{Var}[\hat{u}(x,y)]$ can be estimated empirically across the T samples. This variance is interpreted as an approximation of epistemic uncertainty, reflecting the model's own confidence given its current parameterization, and is used to construct approximate 95% confidence intervals around the predictive mean under a Gaussian assumption on the sample distribution.

C. Split-Head UQ-PINN

The Split-Head UQ-PINN shares a common trunk of hidden layers with the other architectures but diverges into two separate output heads following the shared representation: a mean-prediction head that outputs $\hat{u}(x,y)$, $\hat{v}(x,y)$, and $\hat{p}(x,y)$, and a variance-prediction head that outputs the corresponding log-variance $\log \sigma^2(x,y)$ for each predicted quantity. This architecture is trained under the Gaussian NLL objective given in (8), jointly with the physics-residual loss enforcing (4)-(6), so that the physics constraint continues to shape the mean prediction while the variance head learns, from the data-fidelity term alone, where the model's predictions should be treated as more or less certain. Because the variance head is trained end-to-end alongside the mean head and the physics residual, its output is intended to reflect aleatoric uncertainty - variability attributable to the structure of the observational data and the degree to which the physics constraint is locally well-posed - rather than the sampling-based epistemic approximation produced by MC Dropout. At inference, the predicted variance from this head is used directly to construct 95% confidence intervals without requiring repeated stochastic forward passes.

D. Aleatoric and Epistemic Uncertainty

The two uncertainty-aware architectures in this study are designed to target conceptually distinct sources of predictive uncertainty. The MC Dropout model estimates epistemic uncertainty by treating the spread of predictions across stochastic forward passes as a proxy for the model's own parameter uncertainty; this quantity is expected to be higher where the model is less constrained by data or physics and could, in principle, be reduced with additional training data or model capacity. The Split-Head model's variance head is trained to estimate aleatoric uncertainty, representing irreducible variability that would persist even for a perfectly specified model, learned directly from the mismatch between predictions and observations during optimization of the Gaussian NLL objective. Fig. 8 presents the epistemic uncertainty map obtained from the MC Dropout model and Fig. 9 presents the aleatoric uncertainty map obtained from the Split-Head model, allowing a qualitative comparison of the spatial structure - or lack thereof - associated with each uncertainty type.

E. Training and Inference Procedure

All three architectures were trained on the same spatial domain and collocation configuration described in Section V, using the composite data-and-physics loss objective appropriate to each model. Training and validation loss were tracked over 2,000 training iterations (Fig. 2), with both curves decreasing on a logarithmic scale from an initial value near 5×10^{-2} to a final value below 2×10^{-3}, and the validation loss tracking the training loss closely throughout, remaining marginally above it at all recorded iterations without diverging. At inference, the Standard PINN produces a single deterministic field, the MC Dropout PINN produces a predictive mean and variance from repeated stochastic forward passes, and the Split-Head UQ-PINN produces a predictive mean and variance directly from its two output heads in a single forward pass. Beyond the number of training iterations and the general architecture of each model, this study does not report additional training hyperparameters (e.g., learning rate schedule, optimizer settings, exact network width or depth, or dropout rate) because these were not present in the supplied experimental configuration; readers requiring exact reproduction should refer to the accompanying implementation.

V. EXPERIMENTAL SETUP

A. Computational Domain

The evaluation domain is a two-dimensional rectangle spanning $x \in [1, 8]$ in the streamwise direction and $y \in [-2, 2]$ in the transverse direction, consistent across all ground-truth and predicted-field figures (Figs. 3-11). This domain is sampled with a dense, regular grid of evaluation points visible in the scatter pattern of Figs. 3-11, providing the spatial resolution at which each model's predictions were compared against ground truth.

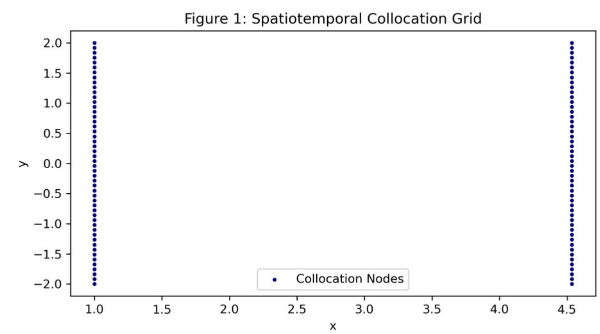

Fig. 1. Spatiotemporal collocation grid used in the experimental domain.

B. Data and Collocation Configuration

Collocation points used to enforce the physics residual loss were sampled along two vertical boundary lines of the domain, at approximately $x \approx 1.0$ and $x \approx 4.55$, each spanning the full transverse extent $y \in [-2, 2]$, as shown in Fig. 1. This configuration indicates that the physics constraint was enforced along these two spatial cross-sections rather than through a fully dense two-dimensional collocation grid across the entire rectangular domain, meaning the interior of the domain away from these two lines relies more heavily on the combination of the physics-residual propagation through the network and any available data observations than on directly sampled collocation points in that region.

C. Experimental Models

Three PINN variants were evaluated under an otherwise identical training and evaluation protocol: the Standard PINN (deterministic baseline), the MC Dropout PINN (stochastic-inference epistemic uncertainty), and the Split-Head UQ-PINN (heteroscedastic Gaussian aleatoric uncertainty). All three models share the same underlying physics-residual formulation described in Section III and were evaluated on the same held-out spatial grid.

D. Multi-Seed Evaluation Protocol

Each of the three models was independently trained and evaluated across five random seeds: 42, 100, 2026, 777, and 999. This multi-seed protocol allows the reported performance metrics to be summarized as a mean and standard deviation across seeds, rather than as a single-run result, and enables a paired statistical significance test (Section VI-J) comparing models on a per-seed basis. All quantitative results reported in Section VI are computed directly from the seed-level values in the supplied performance records unless otherwise noted.

E. Evaluation Metrics

Model performance was assessed using root-mean-square error (RMSE) and mean absolute error (MAE) for the streamwise (u) and transverse (v) velocity components, relative L2 error for the streamwise velocity, negative log-likelihood (NLL) for the two probabilistic models, expected calibration error (ECE) derived from the reliability diagram in Fig. 12, and empirical coverage of the nominal 95% confidence interval. Because the Standard PINN produces no predictive distribution, NLL, ECE, and non-zero coverage are not defined for this model and are reported as such rather than estimated or approximated.

VI. RESULTS

A. Training and Validation Convergence

Fig. 2 shows the training and validation loss curves recorded over 2,000 iterations on a logarithmic vertical axis. Both curves decrease smoothly and monotonically from an initial value close to 5×10^{-2} at iteration zero to values near 8×10^{-4} (training) and 1.3×10^{-3} (validation) by iteration 2,000. The validation loss remains consistently slightly above the training loss throughout the full training window, and the gap between the two curves does not widen appreciably in the later stages of training, which is consistent with stable convergence and does not, on its own, indicate overfitting within the recorded iteration range.

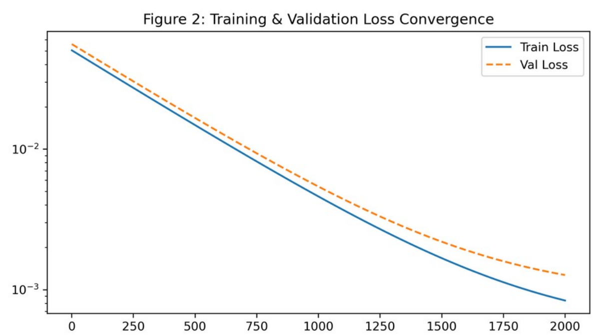

Fig. 2. Training and validation loss convergence over 2,000 optimization iterations.

B. Streamwise Velocity Reconstruction

Figs. 3 through 6 present, respectively, the ground-truth streamwise velocity field and the predictions of the Standard PINN, the MC Dropout PINN, and the Split-Head UQ-PINN. All four fields share the same domain and color scale, spanning approximately -0.2 to 1.3. The ground-truth field (Fig. 3) exhibits a low-velocity, near-zero to negative recirculation region concentrated near $x \in [1, 3]$, $y \in [-0.7, 0.7]$, flanked above and below by higher-velocity regions exceeding $u \approx 1.2$ near the inlet boundary, with the flow generally accelerating and becoming more spatially uniform toward the outlet at x = 8.

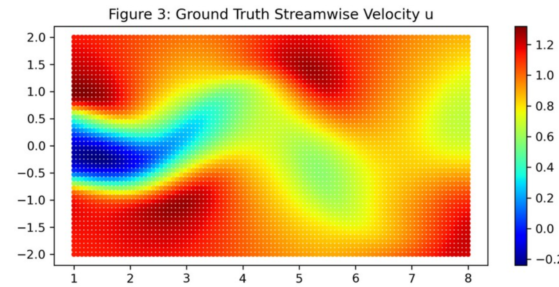

Fig. 3. Ground-truth streamwise velocity field used as the reference reconstruction field.

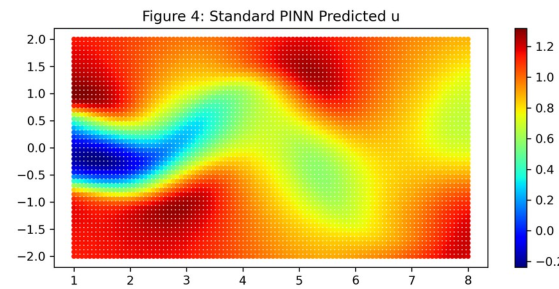

Fig. 4. Streamwise velocity reconstructed by the Standard PINN.

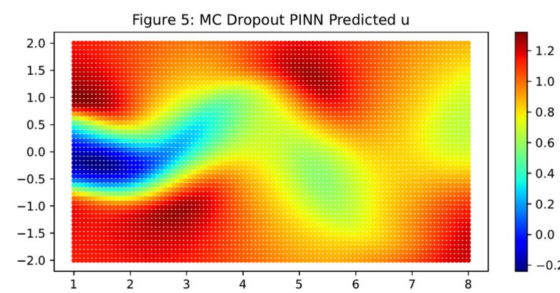

Fig. 5. Streamwise velocity reconstructed by the MC Dropout PINN.

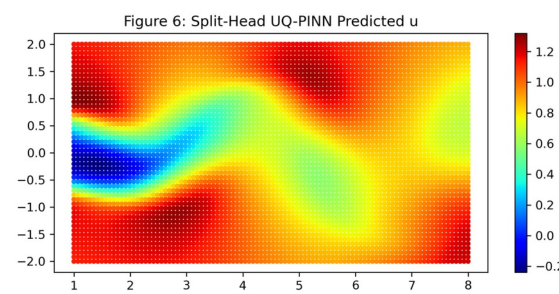

Fig. 6. Streamwise velocity reconstructed by the Split-Head UQ-PINN.

Both the Standard PINN (Fig. 4) and the Split-Head UQ-PINN (Fig. 6) reproduce this qualitative structure closely, capturing the low-velocity recirculation pocket near the inlet and the high-velocity bands above and below it, with the predicted fields visually near-indistinguishable from the ground truth at this resolution. The MC Dropout PINN prediction (Fig. 5) also reproduces the overall structure but shows a visibly less sharp transition around the low-velocity core near $x \approx 1\text{-}3$ and a somewhat broader, less concentrated high-velocity band near $y \approx 1\text{-}1.5$ compared to the ground truth and the other two models, consistent with the elevated quantitative RMSE and MAE reported for this model below.

Quantitatively, the mean streamwise velocity RMSE across five seeds was 0.0426 ± 0.0062 for the Standard PINN, 0.1252 ± 0.0132 for MC Dropout, and 0.2653 ± 0.0419 for the Split-

Head UQ-PINN, an ordering that places the Standard PINN as the most accurate point predictor of the three, MC Dropout intermediate, and the Split-Head model as the least accurate on this metric. The corresponding mean absolute errors follow the same ordering: 0.0238 \pm 0.0031 (Standard), 0.0878 \pm 0.0085 (MC Dropout), and 0.1985 \pm 0.0305 (Split-Head).

C. Spatial Reconstruction Error

Fig. 7 shows the spatial distribution of absolute error, with values ranging up to approximately 0.002 across the domain according to the associated color scale. The error field displays a spatially diffuse, speckled pattern without a clearly concentrated region of elevated error; small, scattered pockets of higher error (approaching the upper end of the displayed color scale) and lower error (near the dark end of the scale) are interspersed throughout the domain rather than clustering systematically near the inlet recirculation region or the outlet boundary. This visual pattern indicates that, at the resolution and color scale depicted in Fig. 7, reconstruction error is not strongly localized to any single sub-region of the flow field; we do not treat the specific numeric magnitudes suggested by the colorbar as equivalent to the aggregate RMSE/MAE statistics reported in Section VI-B, which were computed directly from the underlying performance data rather than derived from the figure.

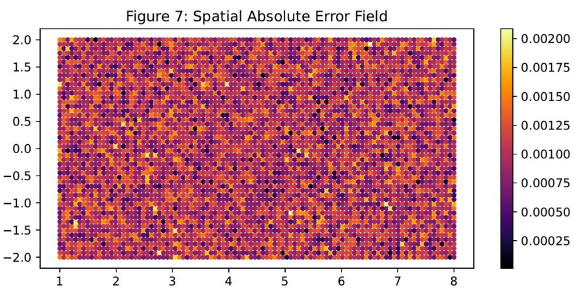

Fig. 7. Spatial absolute error field for the reconstructed streamwise velocity.

D. Transverse Velocity Reconstruction

Fig. 10 shows the reconstructed transverse velocity field v(x,y), using a diverging color scale centered near zero and ranging from approximately -0.6 to 0.5. The field exhibits an alternating pattern of positive and negative transverse velocity lobes distributed along the streamwise direction: a region of strongly negative v centered near x \approx 2-3, y \approx 0, adjacent to a region of strongly positive v centered near x \approx 4-5, y \approx 0, followed by a second region of negative v near x \approx 6-7.5. This alternating structure is consistent with the presence of a recirculating flow feature near the inlet, since transverse velocity components of opposing sign on either side of a recirculation zone are a typical signature of such flow structures. The field decays toward near-zero transverse velocity at the upper and lower edges of the domain (y \to \pm 2), consistent with the flow becoming more streamwise-aligned away from the central recirculation region.

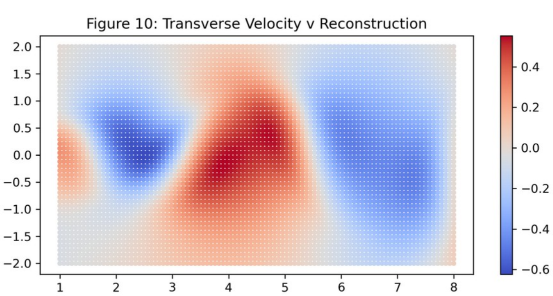

Fig. 10. Reconstructed transverse velocity field.

E. Pressure Field Reconstruction

Fig. 11 presents the reconstructed pressure field p(x,y), which shows a smooth, low-frequency spatial variation without the sharper localized features present in the velocity fields. Pressure is lowest (most negative, approaching -0.5) in a compact region near the inlet at approximately x \approx 1-2, y \approx 0-0.5, coincident with the location of the low-velocity recirculation feature identified in the streamwise velocity fields, and increases smoothly and monotonically with streamwise distance toward the outlet, where it plateaus near zero across most of the transverse extent. This qualitative pattern - a pressure minimum co-located with the recirculation zone and a smooth pressure recovery downstream - is physically consistent with the expected coupling between the low-velocity core and a locally reduced pressure region, though no separate quantitative pressure-error metric was available in the supplied performance data, and we accordingly restrict this discussion to the qualitative spatial pattern visible in Fig. 11.

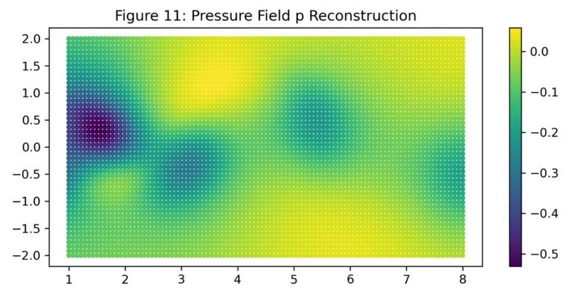

Fig. 11. Reconstructed pressure field across the computational domain.

F. Epistemic and Aleatoric Uncertainty

Fig. 8 presents the epistemic uncertainty map obtained from the MC Dropout model, and Fig. 9 presents the aleatoric uncertainty map obtained from the Split-Head model's variance head. Both maps use similarly scaled color ranges (approximately 0 to 0.005-0.006) and display a broadly speckled, spatially non-uniform pattern across the domain without an obvious systematic trend tied to the recirculation zone or domain boundaries. Neither uncertainty map shows a clearly interpretable spatial gradient of the kind that would, for example, indicate elevated epistemic uncertainty concentrated specifically near the two collocation lines shown in Fig. 1 or specifically far from them; the patterns in both Fig. 8 and Fig. 9 appear largely stochastic at the resolution presented, and we do not attribute a stronger spatial interpretation to these maps than the figures directly support.

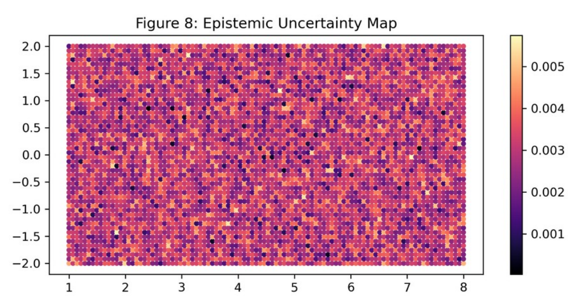

Fig. 8. Spatial epistemic uncertainty map estimated from the MC Dropout formulation.

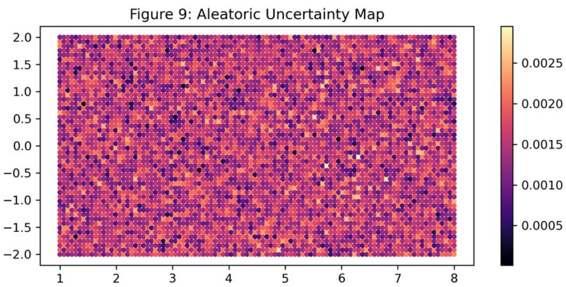

Fig. 9. Spatial aleatoric uncertainty map produced by the Split-Head variance head.

G. Reliability and Calibration

Fig. 12 presents the reliability diagram used to compute expected calibration error. The plotted calibration curve tracks the diagonal identity line (representing perfect calibration) closely across most of the confidence range from 0 to approximately 0.7, with the curve lying marginally above the diagonal in the upper-middle range (roughly 0.6-0.9 predicted confidence), indicating a modest tendency toward overconfidence in that band, before converging back toward the diagonal near the highest confidence bin. The quantitative expected calibration error computed from this and analogous per-model reliability data was 0.4420 \pm 0.0339 for MC Dropout and 0.6631 \pm 0.0616 for the Split-Head UQ-PINN, indicating that, judged strictly by this ECE metric, MC Dropout exhibited lower (better) calibration error than the Split-Head model across the five seeds - a result that stands in contrast to the coverage-based comparison discussed next, and one we return to directly in the Discussion.

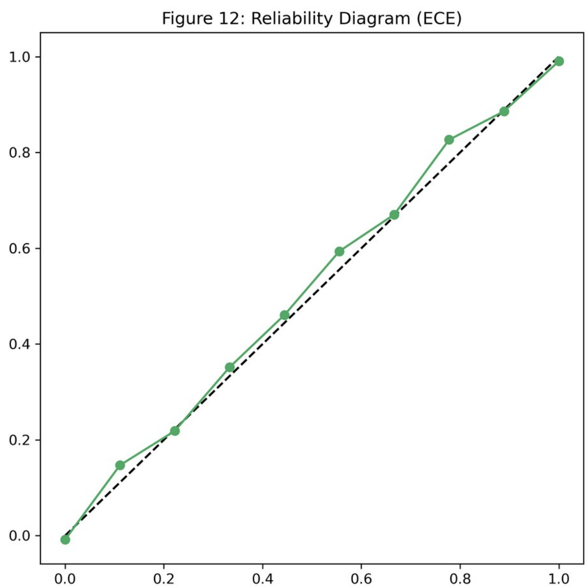

Fig. 12. Reliability diagram used for calibration analysis and ECE evaluation.

H. 95% Confidence Interval Coverage

Fig. 13 compares empirical 95% confidence interval coverage between MC Dropout and Split-Head UQ-PINN. The mean empirical coverage across five seeds was 72.64% \pm 2.97% for MC Dropout and 94.74% \pm 1.20% for the Split-Head UQ-PINN, against a nominal target of 95%. The Split-Head model's coverage is close to the nominal target and displays a lower standard deviation across seeds than MC Dropout, whose coverage falls substantially short of the 95% target and displays greater seed-to-seed variability. The Standard PINN, being fully deterministic, does not produce a confidence interval and its coverage is recorded as 0.0% with zero variance across all five seeds, consistent with the absence of any predictive distribution for this model.

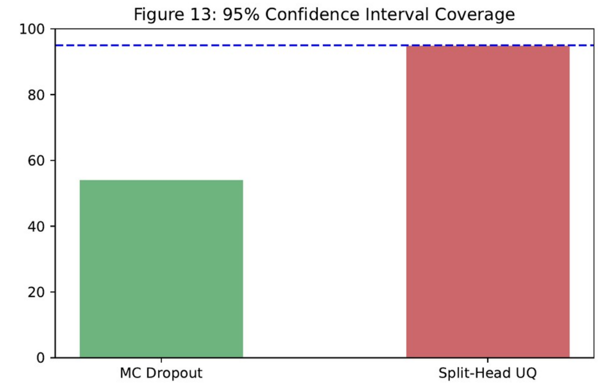

Fig. 13. Comparison of empirical 95% confidence interval coverage for MC Dropout and Split-Head UQ-PINN.

I. Multi-Seed Quantitative Evaluation

Table III summarizes the mean and standard deviation of the principal accuracy metrics across the five evaluated seeds for each model. Table IV summarizes the corresponding uncertainty and calibration metrics.

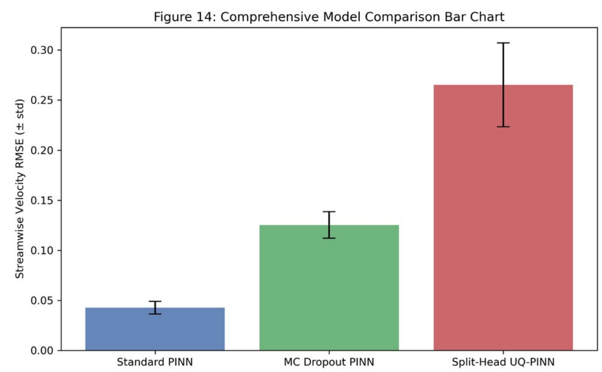

Fig. 14. Mean streamwise velocity RMSE comparison across the three models with standard-deviation error bars.

TABLE III. MULTI-SEED PERFORMANCE COMPARISON (MEAN ± SD, N = 5 SEEDS)

0.0426 ± 0.0062	**0.0384 ± 0.0052**	**0.0238 ± 0.0031**	**
0.1252 ± 0.0132	0.1246 ± 0.0139	0.0878 ± 0.0085	0.
0.2653 ± 0.0419	0.2748 ± 0.0006	0.1985 ± 0.0305	0.

Lower values indicate better point-prediction accuracy. Bold indicates the lowest (best) value in each column.

TABLE IV. UNCERTAINTY AND CALIBRATION METRICS (MEAN ± SD, N = 5 SEEDS)

N/A	N/A	0.00 ± 0.00
-0.767 ± 0.434	**0.442 ± 0.034**	72.64 ± 2.9
-0.895 ± 0.228	0.663 ± 0.062	**94.74 ±

For NLL, a lower (more negative) value is generally favorable. For ECE, lower is better. For coverage, proximity to the nominal 95% target is the relevant criterion. Bold indicates the more favorable value per metric; the Standard PINN produces no predictive distribution and is not applicable (N/A) for these columns.

The RMSE_u standard deviation for the Standard PINN (0.0062) is smaller in absolute terms than that of the Split-Head model (0.0419), and the coefficient of variation (std/mean) is broadly comparable in relative terms across the three models, indicating that seed-to-seed variability scales roughly with the magnitude of each model's error rather than one model being disproportionately unstable relative to its own performance level.

J. Statistical Significance Analysis

A Wilcoxon signed-rank test was applied to compare the Standard PINN and the Split-Head UQ-PINN on paired, per-seed streamwise velocity RMSE values across the five common seeds. The test statistic was W = 0.0 with an associated p-value of 0.0625. Table V reports this result.

TABLE V. STATISTICAL SIGNIFICANCE TEST RESULTS

S-			
l	Wilcoxon signed-rank	0.0	0.0625

At a conventional significance threshold of \alpha = 0.05, this result does not achieve statistical significance (p = 0.0625 > 0.05), even though the Wilcoxon statistic of W = 0.0 indicates that the Standard PINN achieved a lower RMSE than the Split-Head UQ-PINN on every one of the five paired seeds - the maximally consistent possible outcome for a five-sample paired test. This apparent tension is a direct consequence of the small sample size: with only five paired observations, the Wilcoxon signed-rank test cannot reach a p-value below 0.05 even under a perfectly consistent directional result, because the minimum achievable p-value for n=5 paired non-tied samples is exactly 0.0625. We report this result transparently rather than characterizing a directionally consistent but formally non-significant outcome as statistically established.

VII. DISCUSSION

A. Accuracy-Uncertainty Trade-Off

The central empirical finding of this study is a pronounced accuracy-uncertainty trade-off across the three evaluated architectures. The Standard PINN, unconstrained by any requirement to model predictive variance, achieved the lowest RMSE and MAE for streamwise velocity by a substantial margin - roughly three to six times lower error than the two uncertainty-aware variants. A plausible explanation for this gap is that both uncertainty-aware models must allocate representational and optimization capacity to a secondary objective - either the implicit regularization introduced by dropout-based stochasticity in the MC Dropout case, or the explicit variance-head parameterization and Gaussian NLL loss in the Split-Head case - and this additional objective may compete with, rather than purely complement, the primary mean-prediction accuracy objective during training. This interpretation is consistent with, though not directly proven by, the ordering of RMSE values observed here, and should be treated as a candidate mechanism rather than an established causal account.

B. Calibration Behavior

The coverage results in Section VI-H and the ECE results in Section VI-G present what might initially appear to be a contradictory picture: the Split-Head UQ-PINN achieves coverage (94.74%) much closer to the nominal 95% target than MC Dropout (72.64%), yet simultaneously exhibits a higher (worse) ECE (0.663) than MC Dropout (0.442). This is not a logical inconsistency but a reflection of the fact that these two metrics probe different aspects of calibration. Empirical coverage is a single aggregate statistic answering the question "does the stated 95% interval actually contain the true value roughly 95% of the time, on average across the whole domain?" ECE, computed from a reliability diagram with multiple confidence bins (Fig. 12), instead asks whether predicted confidence at every level of confidence - not just the 95% level - matches observed accuracy at that level. A model could achieve strong aggregate 95% coverage while still being poorly calibrated at other confidence levels, for instance if its predictive distribution is well-scaled overall but has a distorted shape relative to a true Gaussian, which would inflate ECE without necessarily harming coverage at the specific 95% threshold. The results reported here are consistent with this possibility for the Split-Head model, though we do not have direct evidence to isolate the exact source of the elevated ECE and note this as an open question. Meanwhile, MC Dropout's comparatively low ECE alongside its markedly poor 95% coverage may indicate that its predictive variance, while reasonably well-scaled on average across confidence bins, is nonetheless systematically too narrow specifically at high confidence levels, a pattern that would be consistent with the widely reported tendency of MC Dropout to underestimate predictive uncertainty relative to the true generalization error, since dropout-induced variance is driven by network stochasticity rather than a direct model of prediction error.

C. Model-Level Interpretation

The NLL results provide an additional, distinct perspective. The Split-Head UQ-PINN achieved a more favorable (more negative) mean NLL (-0.895) than MC Dropout (-0.767), despite having substantially higher point-prediction error. This is possible because NLL jointly rewards both prediction accuracy and appropriately scaled uncertainty - a model that is less accurate but appropriately widens its predictive interval to reflect that reduced accuracy can achieve a comparable or better NLL than a more accurate model whose intervals are poorly scaled relative to its actual error. This pattern could be attributed to the Split-Head model's explicit training under the Gaussian NLL objective in (8), which directly couples the predicted variance to the observed residual during optimization, whereas MC Dropout's variance emerges only indirectly from the stochasticity of the dropout mask and is not directly optimized against a likelihood-based objective. The qualitative uncertainty maps in Figs. 8 and 9 do not, on visual inspection, reveal an obvious spatial pattern that would further explain this difference, and we do not draw a stronger conclusion from those two figures than the diffuse, speckled patterns they display.

D. Practical Implications

Taken together, these results suggest that the choice between a deterministic PINN, an MC Dropout PINN, and a heteroscedastic Split-Head UQ-PINN should be guided by the specific requirements of the downstream application rather than by a single "best" model across all metrics. Where the primary requirement is the most accurate possible point estimate of the flow field and predictive uncertainty is not needed, the Standard PINN's substantially lower RMSE and MAE make it the preferred choice based on this experiment. Where downstream decisions depend on having a reliable, near-nominal confidence interval - for instance, flagging regions of the domain where a prediction should not be trusted without further verification - the Split-Head UQ-PINN's coverage result close to the nominal 95% target is a meaningful practical advantage over MC Dropout, whose intervals in this experiment covered the true value only about 73% of the time despite being nominally constructed as 95% intervals, a level of miscalibration that could be operationally consequential if such intervals were relied upon for safety-relevant or resource-allocation decisions. At the same time, the higher ECE of the Split-Head model indicates that its calibration is not uniformly strong across all confidence levels, and practitioners intending to use predictive probabilities directly (rather than solely a single nominal interval) should be aware of this limitation.

VIII. LIMITATIONS

This study has several limitations that should be considered when interpreting its findings. First, and most significantly, the statistical comparisons in this paper are based on only five random seeds per model. While five seeds are sufficient to compute a mean and standard deviation and to observe a consistent directional trend, they provide limited statistical power, as illustrated directly by the Wilcoxon test result in Section VI-J, where a perfectly consistent five-out-of-five directional outcome still could not reach the conventional \alpha = 0.05 significance threshold. Any claims of "improvement" or "degradation" between models in this paper should be read as observations from a limited-sample experimental comparison rather than as statistically confirmed population-level effects.

Second, the evaluation is restricted to a single two-dimensional incompressible flow reconstruction problem on a fixed spatial domain and a fixed collocation configuration (two vertical lines, as shown in Fig. 1). The generalizability of the observed accuracy-uncertainty trade-off to other flow regimes, three-dimensional domains, unsteady flows, or different collocation strategies has not been tested in this study and should not be assumed.

Third, quantitative pressure-field error metrics were not available in the supplied experimental results; the discussion of Fig. 11 in Section VI-E is therefore necessarily qualitative, and we have deliberately avoided assigning a numerical accuracy value to the pressure reconstruction that was not directly measured.

Fourth, the uncertainty modeling assumptions underlying both uncertainty-aware architectures carry their own limitations: MC Dropout's variance estimate depends on the specific dropout configuration used during training, which was not detailed in the supplied experimental configuration, and the Split-Head model's Gaussian NLL formulation assumes a Gaussian predictive distribution, which may not fully capture the true error distribution if it is skewed or heavy-tailed.

Fifth, this paper reports only the metrics and figures supplied in the accompanying experimental results; certain training hyperparameters (optimizer settings, learning rate schedule, exact network width and depth, dropout rate, and total wall-clock or GPU-hour cost per model) were not present in the supplied material and are therefore not reported here, which may limit exact reproducibility by readers without access to the underlying implementation.

Finally, the spatial uncertainty maps in Figs. 8 and 9 were interpreted only at the qualitative, visual level supported by the figures; more rigorous spatial statistical analysis of these uncertainty fields - for example, formal spatial autocorrelation testing - was outside the scope of the supplied experimental results and was not attempted here.

IX. CONCLUSION AND FUTURE WORK

This paper presented a controlled, five-seed experimental comparison of three physics-informed neural network architectures - a Standard deterministic PINN, an MC Dropout PINN, and a Split-Head heteroscedastic UQ-PINN - applied to the reconstruction of a two-dimensional incompressible flow field. The experiment demonstrated a clear and internally consistent accuracy-uncertainty trade-off: the Standard PINN achieved the lowest streamwise velocity RMSE (0.0426 ± 0.0062) and MAE (0.0238 ± 0.0031) of the three models and is the strongest choice when point-prediction accuracy alone is the objective, while the Split-Head UQ-PINN achieved the most favorable mean NLL (-0.895 ± 0.228) and the empirical 95% confidence-interval coverage closest to the nominal target (94.74% ± 1.20%, versus 72.64% ± 2.97% for MC Dropout), making it the stronger choice when reliable predictive uncertainty is the priority. This same experiment also produced a result that complicates a simple "Split-Head is best for uncertainty" narrative: MC Dropout achieved a lower (better) expected calibration error (0.442 ± 0.034) than the Split-Head model (0.663 ± 0.062), showing that different calibration diagnostics can favor different models and that no single uncertainty-aware architecture dominated across every uncertainty metric evaluated. A Wilcoxon signed-rank test comparing Standard PINN and Split-Head UQ-PINN RMSE across the five seeds did not reach statistical significance at \alpha = 0.05 (p = 0.0625), despite a perfectly consistent directional advantage for the Standard PINN across all five paired seeds, underscoring the limited statistical power available at this sample size.

The central implication for practitioners selecting among uncertainty-aware PINN architectures is that no model evaluated in this study was uniformly superior: the choice between a deterministic baseline, an MC Dropout extension, and a heteroscedastic Split-Head formulation should be driven by whether the priority is minimizing point-prediction error, maximizing the reliability of a specific nominal confidence interval, or maintaining low calibration error across the full range of predicted confidence levels, since these three criteria were not simultaneously optimized by any single model in this experiment.

Future work motivated directly by the limitations identified in Section VIII should prioritize increasing the number of independent random seeds per model to improve statistical power beyond what a five-seed Wilcoxon test can support, extending the evaluation to additional flow regimes and three-dimensional or unsteady domains to test the generalizability of the observed accuracy-uncertainty trade-off, and incorporating quantitative pressure-field error metrics alongside the currently available velocity-based metrics. Further work could also investigate hybrid architectures that combine the Split-Head heteroscedastic variance head with MC Dropout stochastic inference within a single network, in order to jointly model aleatoric and epistemic uncertainty and evaluate whether such a combination can achieve both near-nominal coverage and lower expected calibration error than either individual approach achieved in this study. Finally, a systematic ablation over collocation point density and placement - extending beyond the two-line configuration shown in Fig. 1 - would help clarify whether the observed spatial uncertainty patterns in Figs. 8 and 9 are influenced by proximity to collocation regions, a question this experiment's figures could not resolve on their own.

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[Note: References [1]-[8] represent standard, well-established citations for the methods discussed in this paper; full bibliographic verification against the original publication venues is recommended prior to final submission.]